PS-15: Skin Lesion Classification (HAM10000)

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GitHub:https://github.com/AnishIdhayan-1412/DL-CBL-PS15

1. Problem and Objective

Skin cancer is easier to treat when found early, and dermatoscopic images can support screening. The task is to classify a dermatoscopic image into one of seven pigmented lesion types. Melanocytic nevi are about 67% of the data, so accuracy is misleading. Success is judged by **macro F1** on a held-out test set, together with balanced accuracy and melanoma recall.

2. Dataset and Preprocessing

HAM10000 (Harvard Dataverse), 10,015 images, 7 classes; the full set was used. The images belong to 7,470 unique lesions and 1,956 lesions have more than one image, so a random split would leak near-identical images. I split by **lesion_id** (StratifiedGroupKFold, seed 42) into train 6,691 / validation 1,674 / test 1,650 images and verified that no lesion appears in two splits. Images were resized to 224 x 224 and normalised with ImageNet statistics. Training augmentation: random resized crop, horizontal and vertical flips, rotation (20 deg) and colour jitter; validation and test data were not augmented. Imbalance was handled with a class-weighted cross-entropy loss (square-root inverse frequency; weights range from 0.46 for nevus to 3.81 for dermatofibroma).

3. Methodology

Two ImageNet-pretrained CNNs, ResNet18 and EfficientNet-B0, were fine-tuned end to end with a new 7-class output layer. Both were compared on the validation set only; the selected model was evaluated once on the test set. Training used mixed precision on a Colab T4 GPU, a cosine learning-rate schedule, and early stopping on validation macro F1 (the best checkpoint was kept). Python 3, PyTorch 2.11, torchvision 0.26, scikit-learn 1.6.1.

Hyperparameter	Value	Hyperparameter	Value
Optimiser	AdamW (wd 1e-4)	Batch size	64
Learning rate	3e-4, cosine decay	Max epochs / patience	12 / 4
Loss	Weighted cross-entropy	Input size / seed	224 x 224 / 42

Table 1: Training hyperparameters.

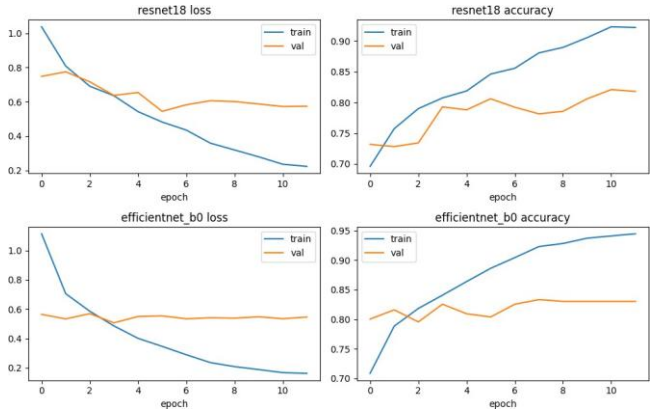
4. Results

Model	Split	Accuracy	Macro F1	Balanced acc.	Melanoma recall	Train time
ResNet18 (selected)	Test	0.82	0.68	0.66	0.56	604 s
ResNet18	Validation	0.82	0.70 (best)	-	-	604 s
EfficientNet-B0	Validation	0.83	0.70 (best)	-	-	639 s

Table 2: Overall results. Test numbers are from the 1,650 held-out test images; validation numbers were used only to choose the model.

Class (test)	akiec	bcc	bkl	df	mel	nv	vasc
Precision	0.63	0.78	0.71	0.71	0.49	0.91	0.79
Recall	0.41	0.76	0.66	0.44	0.56	0.92	0.86
F1	0.50	0.77	0.68	0.55	0.52	0.91	0.83
Support	41	85	184	27	187	1104	22

Table 3: Per-class test metrics for ResNet18.



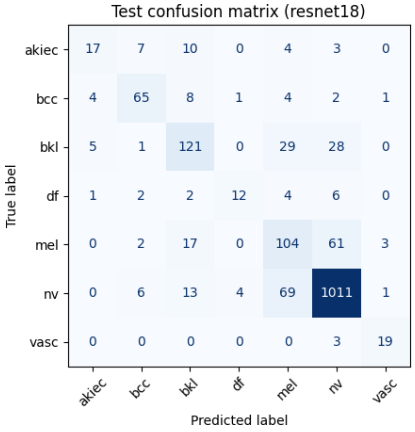


Fig. 1: Learning curves (ResNet18 top, EfficientNet-B0 bottom). Fig. 2: Test confusion matrix (ResNet18).

5. Analysis and Discussion

Confusions. The most frequent errors are nevus predicted as melanoma (69) and melanoma predicted as nevus (61), followed by benign keratosis predicted as melanoma (29) or nevus (28). These lesions are pigmented and visually similar. Actinic keratosis (recall 0.41) and dermatofibroma (0.44) are weak because they have few training images (238 and 66). **Over-fitting.** ResNet18 reaches about 92% training accuracy against 82% validation accuracy, and validation loss stays near 0.57 while training loss keeps falling, so there is mild over-fitting. Augmentation, weight decay and early stopping reduced it but did not remove it. **Misclassified examples** (8 inspected): a pale scarred basal cell carcinoma with hair was predicted as dermatofibroma; a faint, diffuse actinic keratosis was predicted as benign keratosis; an irregular brown nevus and a small dark nevus were predicted as melanoma; a smooth brown benign keratosis was predicted as nevus; a benign keratosis with a dark centre was predicted as melanoma; a melanoma with unusual blue-red colour and lighting was predicted as vascular lesion (low confidence 0.43); and a flat streaked melanoma was predicted as nevus with 0.99 confidence. **Specific challenge.** Leakage was prevented by the lesion-level split (zero overlap verified), and imbalance by weighted loss and macro F1 reporting. Grad-CAM on the 8 errors shows the model focuses on the lesion and not on rulers, hair or image borders, so there is no sign of shortcut learning; the maps are fairly centre-heavy because lesions are usually centred. The two backbones tied on validation macro F1 (0.70), so ResNet18, the smaller model, was selected. Validation scores for small classes rest on only 20-48 images and are noisy.

6. Real-world Relevance, Limitations and Ethics

A model like this could help triage lesions in clinics with few dermatologists. It is **not** ready for deployment: it misses 44% of melanomas, and a missed melanoma is far costlier than a false alarm. HAM10000 comes from two sites and mostly shows light skin tones, so performance on other skin types and phone cameras is unknown and may be biased. Patient images are sensitive data, and any tool should support a clinician and not replace one.

7. Conclusion and Future Work

A fine-tuned ResNet18 with a lesion-level split reached 0.82 test accuracy and 0.68 macro F1, with melanoma recall of 0.56. Next, I would raise melanoma recall by lowering its decision threshold and using focal loss, and test on an external dataset such as ISIC 2019.

Declaration: An AI assistant (Claude) helped generate the notebook code and draft this report; all results come from my own run. **Dataset:** Tschandl P., Rosendahl C., Kittler H., "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions", Sci. Data 5:180161 (2018), doi:10.7910/DVN/DBW86T.